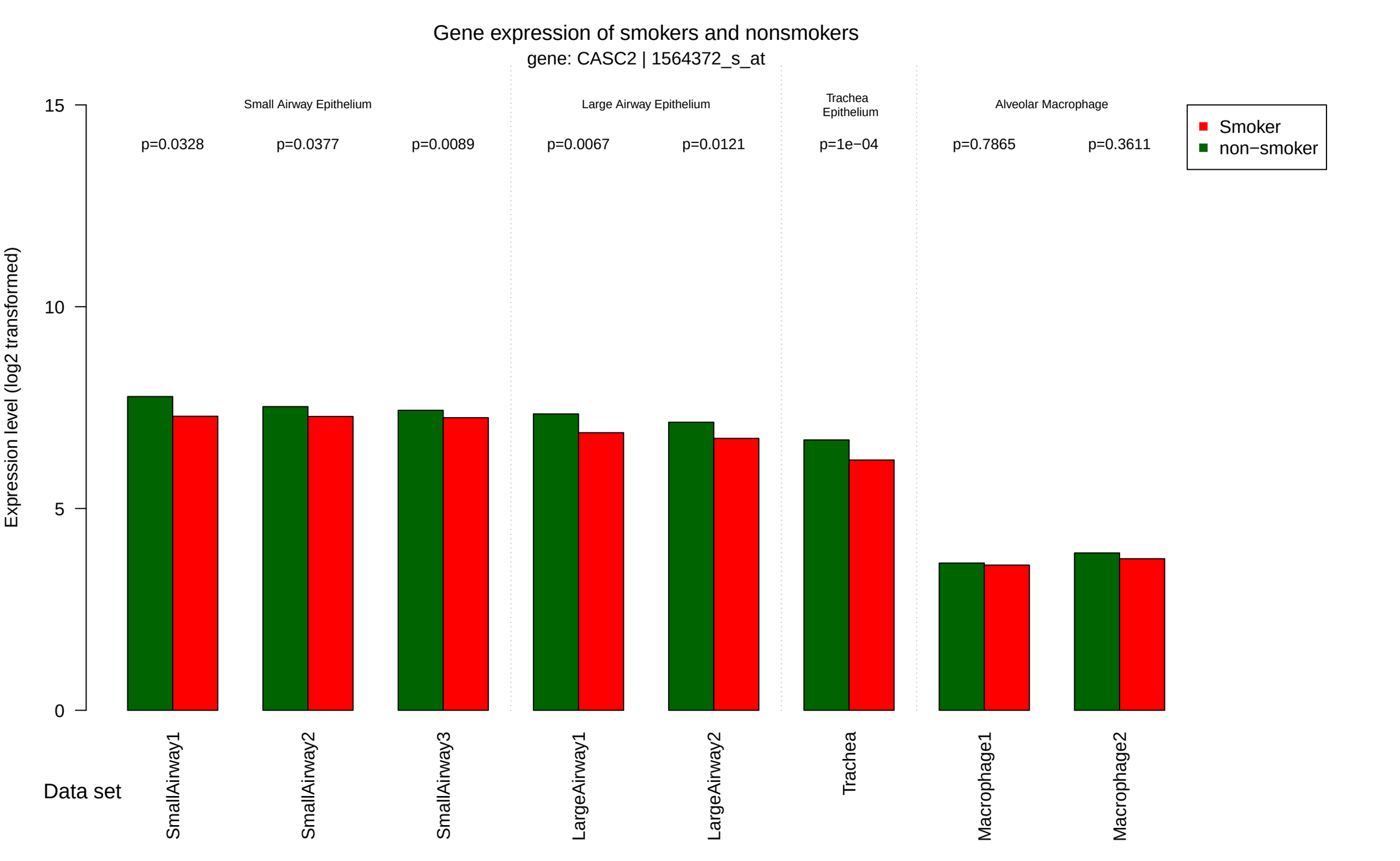
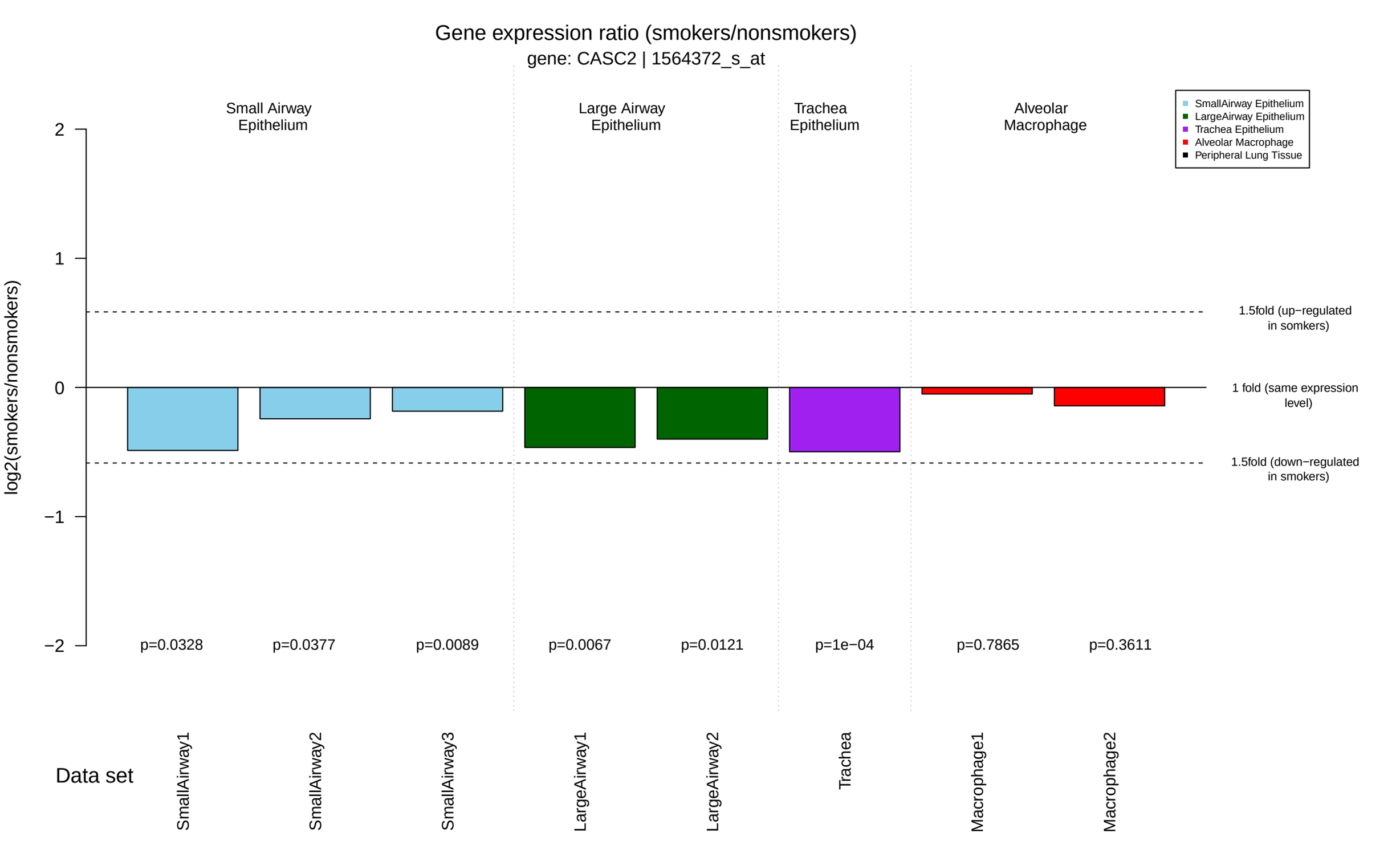
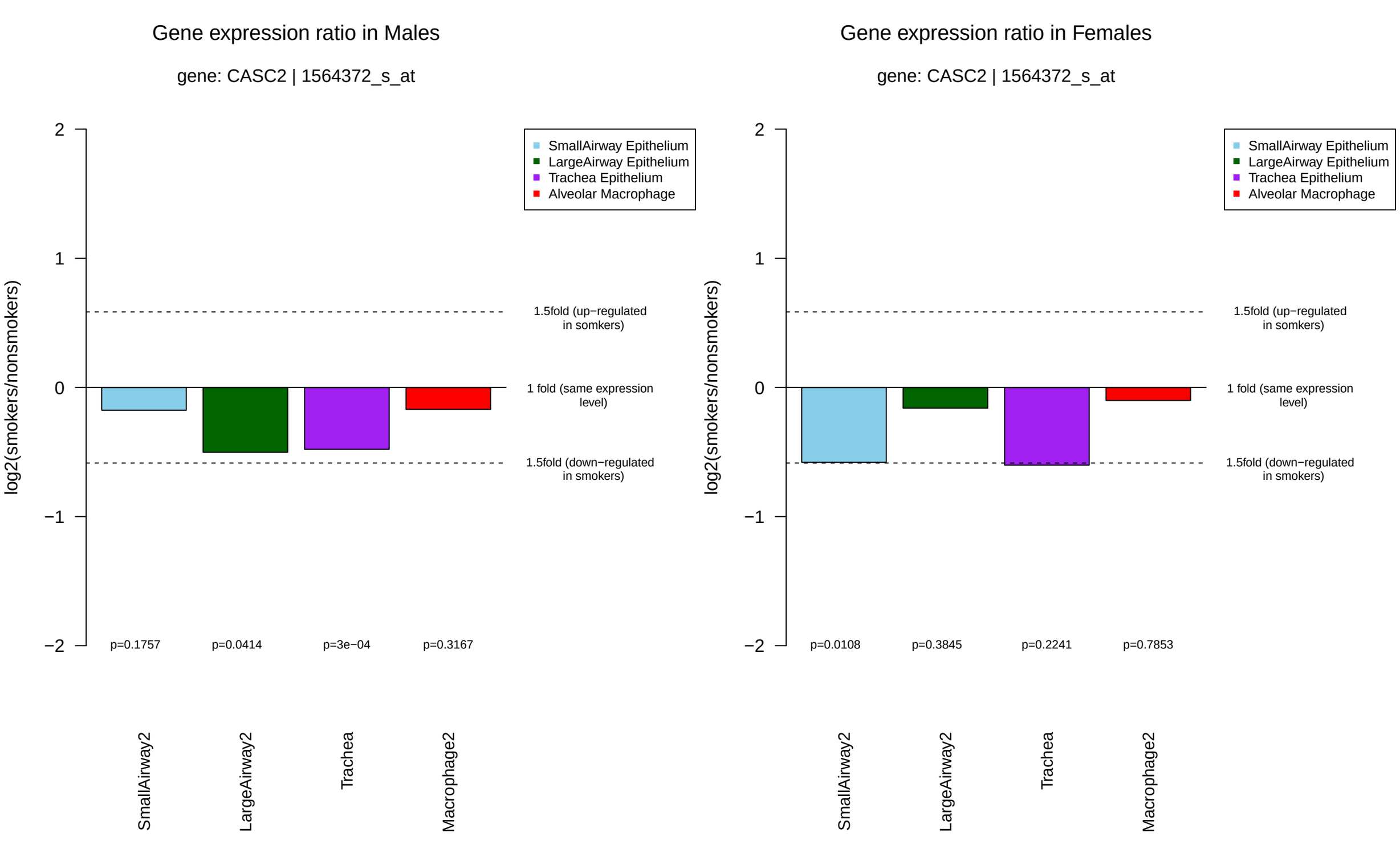
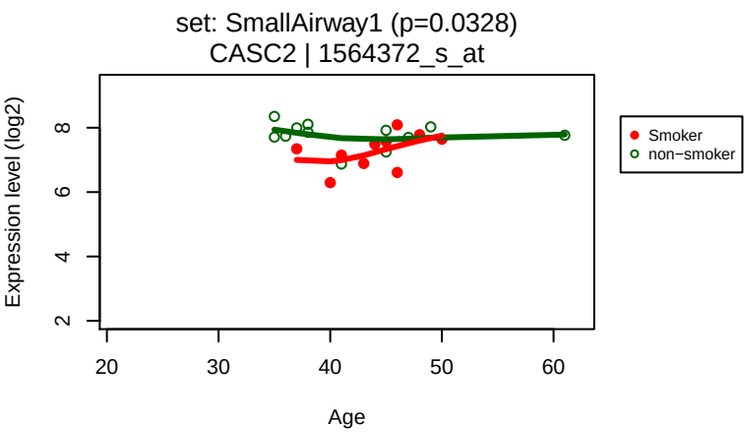
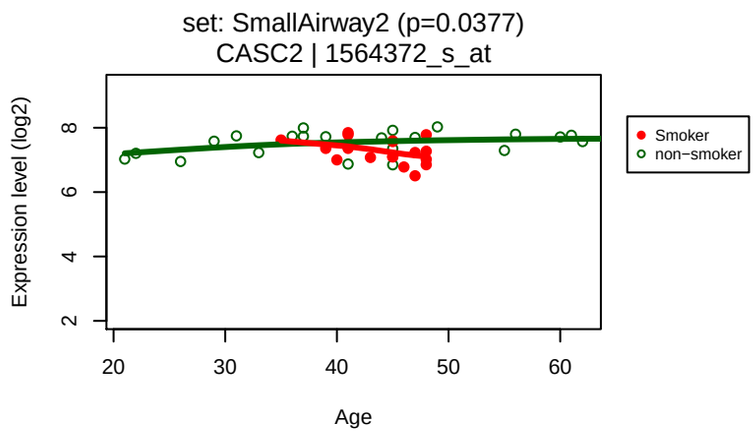
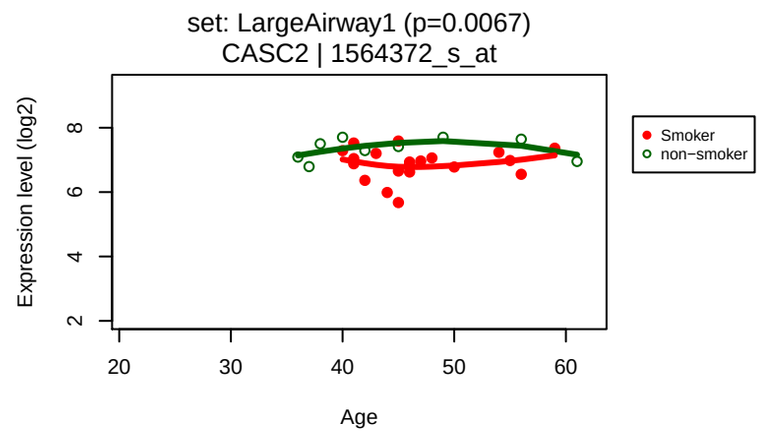

Gene expression of smokers and nonsmokers
gene: CASC2 | 1564372_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0328
p=0.0377
p=0.0089
p=0.0067
p=0.0121
p=1e−04
p=0.7865
p=0.3611
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

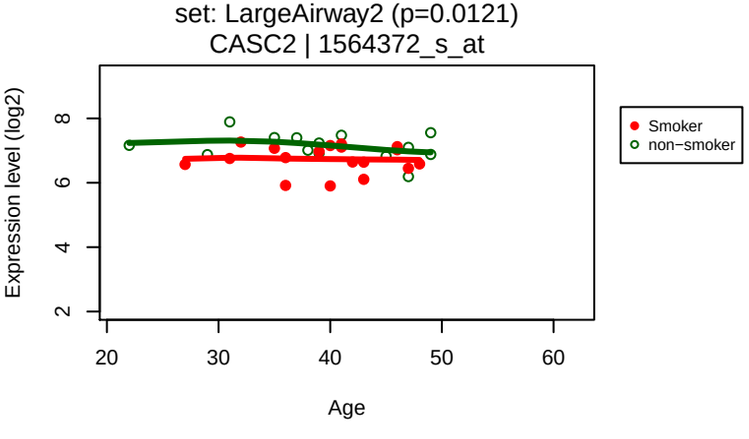

Gene expression ratio (smokers/nonsmokers)
gene: CASC2 | 1564372_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0328
p=0.0377
p=0.0089
p=0.0067
p=0.0121
p=1e−04
p=0.7865
p=0.3611
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

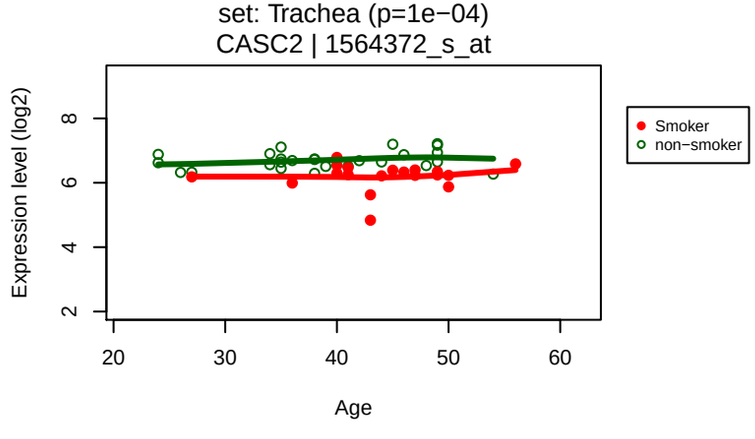

Gene expression ratio in Males
gene: CASC2 | 1564372_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1757
p=0.0414
p=3e−04
p=0.3167
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CASC2 | 1564372_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0108
p=0.3845
p=0.2241
p=0.7853
SmallAirway2
LargeAirway2
Trachea
Macrophage2

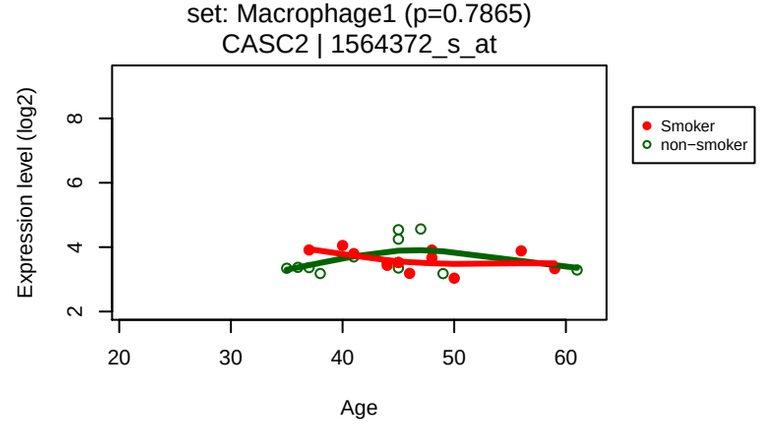

set: SmallAirway1 (p=0.0328)
CASC2 | 1564372_s_at
Expression level (log2)
Age
Smoker
non−smoker

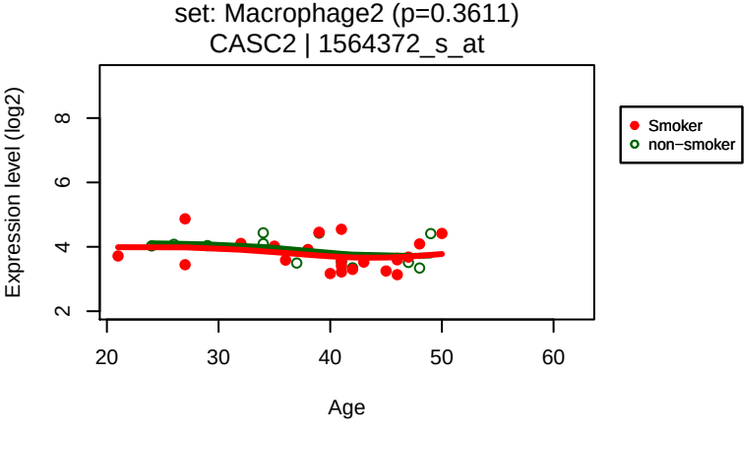

set: SmallAirway2 (p=0.0377)
CASC2 | 1564372_s_at
Expression level (log2)
Age
Smoker
non−smoker

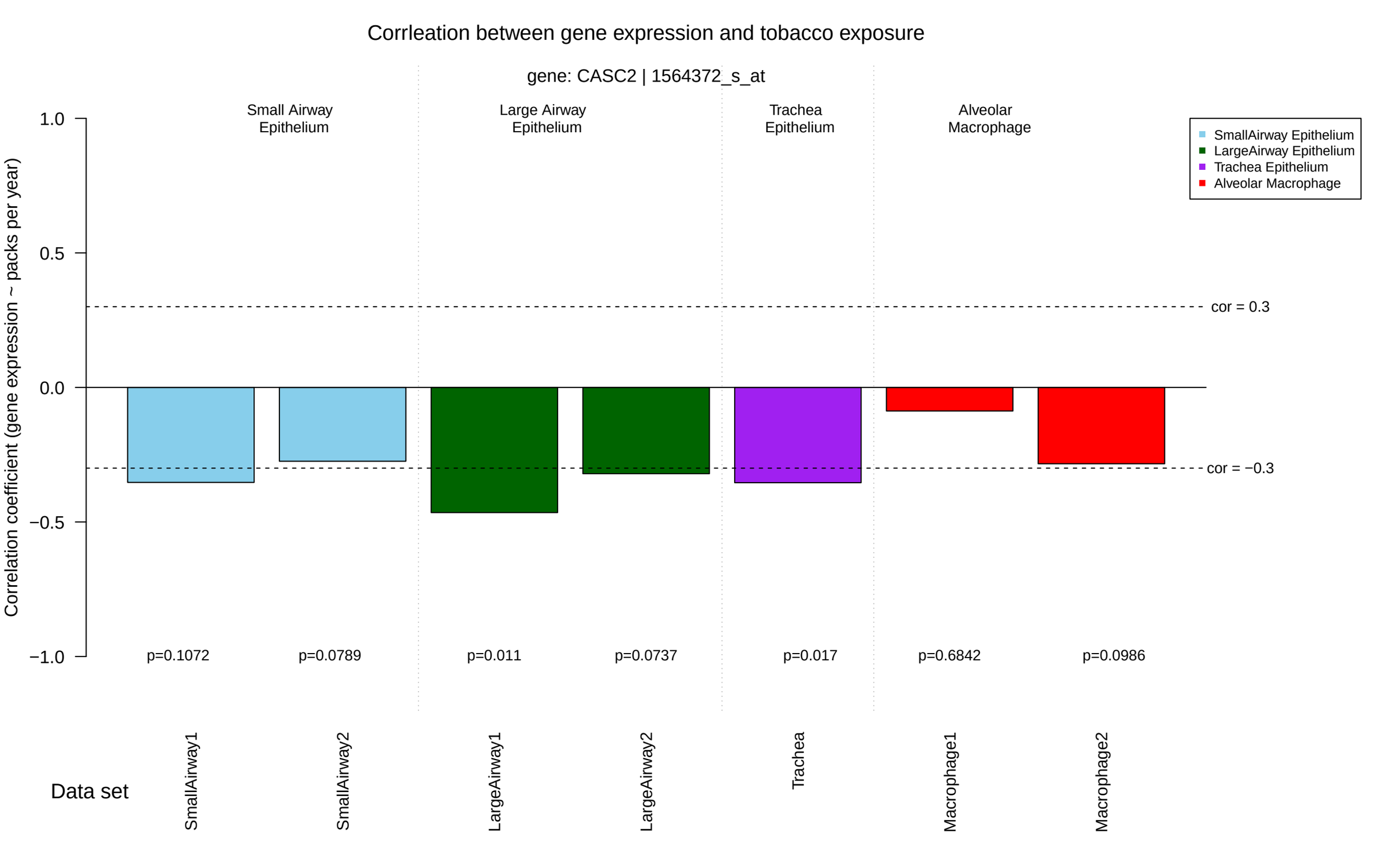

set: LargeAirway1 (p=0.0067)
CASC2 | 1564372_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0121)
CASC2 | 1564372_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
CASC2 | 1564372_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7865)
CASC2 | 1564372_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3611)
CASC2 | 1564372_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CASC2 | 1564372_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1072
p=0.0789
p=0.011
p=0.0737
p=0.017
p=0.6842
p=0.0986
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2